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Experiment No.: 01
Experiment Name: Regression and Classification on WEKA
Date: 19/09/25

Objective:

To implement and analyze regression (Linear Regression on CPU dataset) and classification (Decision Tree models on Iris dataset) using Weka and compare their performance.

Theory/Concepts:

- **Regression:** A type of supervised learning aimed at forecasting continuous numerical outputs.
  - *Linear Regression* constructs a predictive model by assigning weights to input features, optimizing to reduce prediction errors.
  - Model effectiveness is measured through metrics such as the correlation coefficient, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and relative error percentages.
- **Classification:** A supervised learning approach used to categorize data into predefined classes.
  - *DecisionStump* is a simple classifier that makes predictions based on a single attribute split.
  - *RandomTree* builds decision trees by randomly choosing attributes at each split, enhancing model diversity and resilience.
  - Evaluation involves metrics like classification accuracy, confusion matrices, and error rate analysis.

Procedure:

1. Open Weka Explorer.

2. Load dataset:

3. CPU dataset for regression.

4. Iris dataset for classification.

For regression:

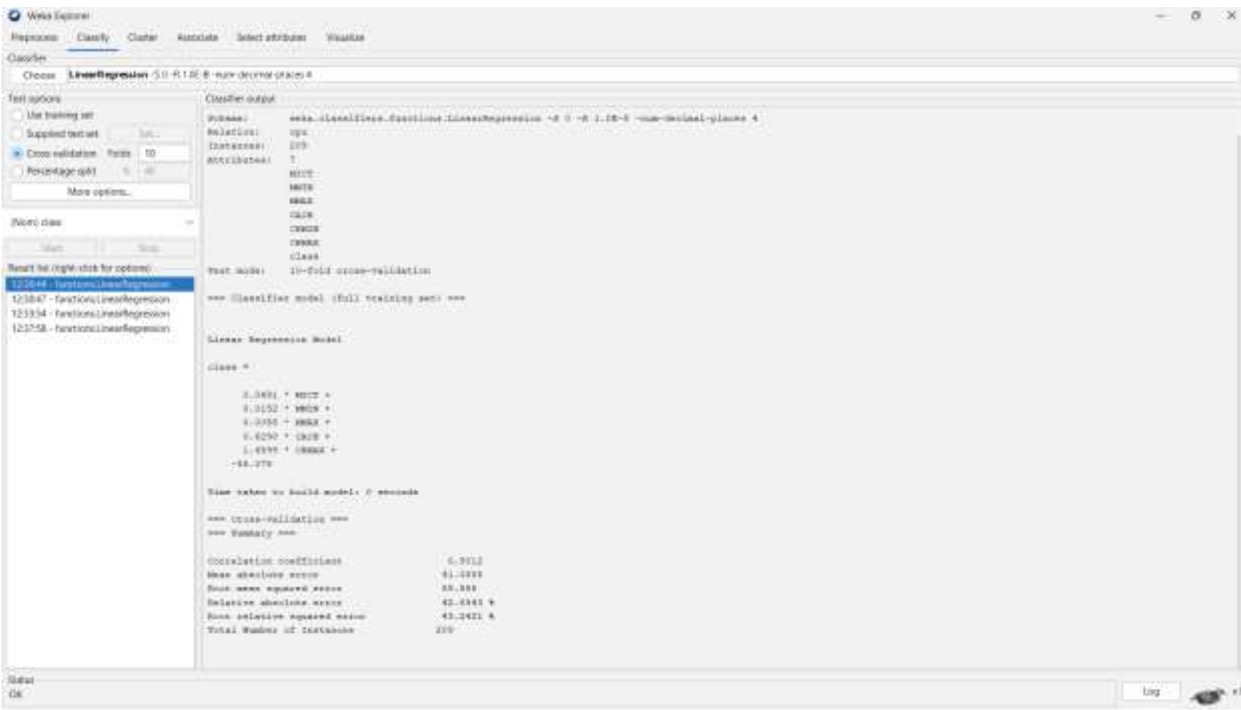
- Select Functions → LinearRegression.
- Use test options: 10-fold cross-validation and 66% percentage split.

For classification:

- Select classifiers: trees.DecisionStump and trees.RandomTree.
- Use 10-fold cross-validation.

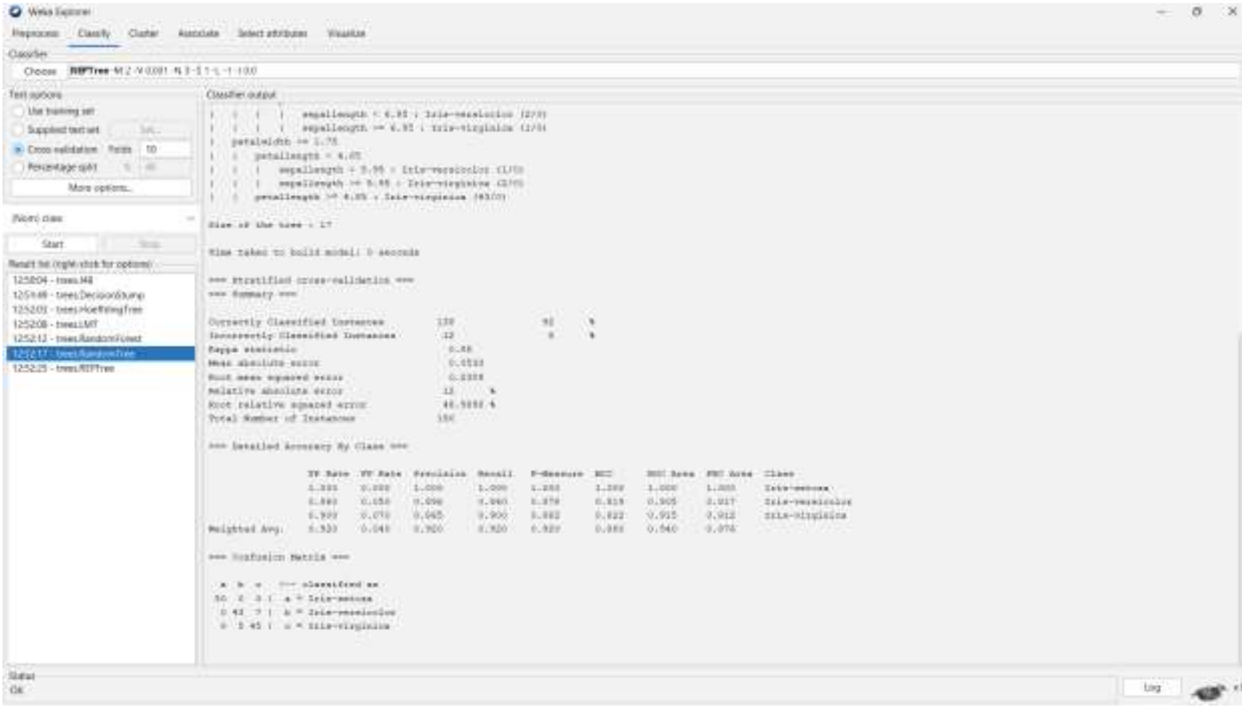
Run the models and note performance metrics.

Output:




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
LinearRegression
Test options
Use training set
Supplied test set
Cross-validation
Folds
10
Percentage split
More options...
Classifier output
Result list (right-click for options)
Status
OK
Log


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
REPTree
Test options
Use training set
Supplied test set
Cross-validation
Folds
10
Percentage split
More options...
Start
Classifier output
Result list (right-click for options)
Status
OK
Log



Result:

Linear Regression delivered reliable predictive performance on the regression task, with correlation coefficients ranging from 0.90 to 0.92.
In the classification domain, RandomTree demonstrated superior results compared to DecisionStump, reaching an accuracy of 92%, while DecisionStump managed 67%.
Overall, Weka proves to be a powerful platform for implementing and evaluating both regression and classification models effectively